

# ASHWIN SATISH

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## EDUCATION

Northeastern University - Khoury College of Computer Science

Sept 2024 - Dec 2026 | Boston, MA

*Master's of Science, Data Science*

GPA: 3.83/4

Machine Learning, NLP, MLOps, Algorithms, Data Mining, Data Science Capstone

Christ University

June 2020 - May 2023 | Bangalore, India

*Bachelor's of Science, Computer Science*

GPA: 3.53/4

## EXPERIENCE

Gyori Lab for Computational Biomedicine | [Machine Learning Intern](#)

Jan 2026 - Present | Boston, MA

- Engineered [TrialSynth](#)'s clinical trial LLM extraction pipeline by parsing **35M** PubMed records, identifying **278K+** **publications** linked to registered trials, transforming **13K+** **papers** into structured datasets for knowledge graph integration.
- Designed & implemented [CoGEx](#)'s **Trial Result** schema, converting trial datasets to **7 Neo4j node** types & **9 relationship** types, generating **115K+ nodes**, **116K+ relationships** alongside **13 REST APIs** & a [Flask search interface](#) enabling queryable clinical trials within the knowledge graph.
- Boosted adverse event entity coverage from **32.5%** to **66.8%** across **13k+** clinical trial papers by building a multi-ontology extraction pipeline with systematic noise filtering.

Northeastern University | [Machine Learning Researcher](#)

Jan 2025 - Dec 2025 | Boston, MA

- Boosted true positives from **2% to 77% (maintaining zero false positives)** for Lyme cell detection via 7-model CNN ensemble.
- Built a pipeline for reconstruction & masking of **30,000+ cell images**, saving the biology team weeks of manual work.
- Improved accuracy by **13%** via augmentation & L2/dropout regularization, achieving production-ready performance.

Northeastern University | [Teaching Assistant \(Foundations of Data Science\)](#)

Aug 2025 - Dec 2025 | Boston, MA

- Mentored **300+ students** in linear algebra & statistics for ML & Data Science via problem-solving sessions & **NumPy** demos.
- Conducted **4+ weekly** office hours clarifying topics like constraint optimization, low-rank approximation, & regression.

CarerAI | [Data Science Intern](#)

Jul 2023 - May 2024 | Gurgaon, India

- Prevented **2,000+ unnecessary procedures** for customers by uncovering **500+** surgery avoidance cases through KNN analysis of patient records against historical data.
- Leveraged EHR for a **risk-score**, improving classification accuracy by **26.4%**, increasing confidence in surgery avoidance cases.
- Improved KNN's accuracy by **20%** using DBSCAN to remove record outliers, reducing false positives in surgery recommendations.

## SKILLS

**Programming** | **Data:** Python, SQL, R, Pandas, NumPy, PySpark, Scikit-learn, TensorFlow, PyTorch, Hugging Face, OpenCV

**Visualization** | **Cloud:** Git, Tableau, Matplotlib, Seaborn, AWS, GCP, Azure, PostgreSQL, Snowflake, Docker, Neo4j

**AI** | **Modeling:** Transformers, BERT, LoRA, AutoML, T5, LangChain, LangGraph, Kubeflow, Ollama, PydanticAI, MLFlow

## PROJECTS

AI-Powered Pokémon Battle Strategy Engine | [Github](#)

- Built & dockerized hybrid **Multi-Agent System (MAS)** integrating **Llama 3.2** with deterministic backend for strategy analysis with a recommendation system for optimal team synergies by ingesting **1M+ aggregated usage stats** via an ETL pipeline.
- Engineered a RAG pipeline via ChromaDB & Sentence-Transformers, **indexing 40,000+ strategy docs** to ground LLM inference in semantic context.

PantryPilot (2nd Place | MLOps Innovation Expo @ Google) | [Github](#)

- Contributed to the fine tuning, evaluation & automated deployment (**GCP**) of **Llama 3.2 3B** for personalized inventory recipes, resulting in **50% reduction** in **dietary violations** via LoRA adaptation.
- Collaborated on DPO feedback with Teacher model (**Llama 3 70B**) for preference labelling, VLM for OCR & Veo 3.1 for video.